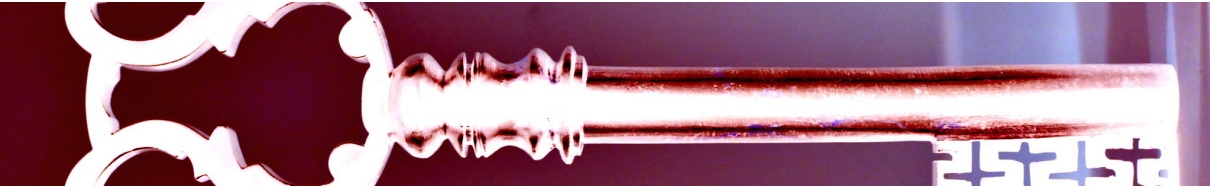
universität innsbruck



Symmetric Encryption

Cryptography

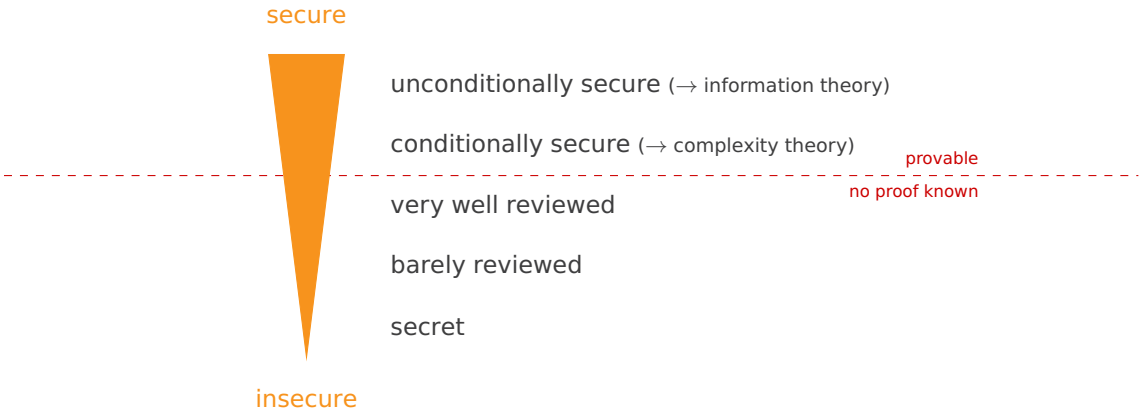
Rainer Böhme

19 March 2026

Agenda

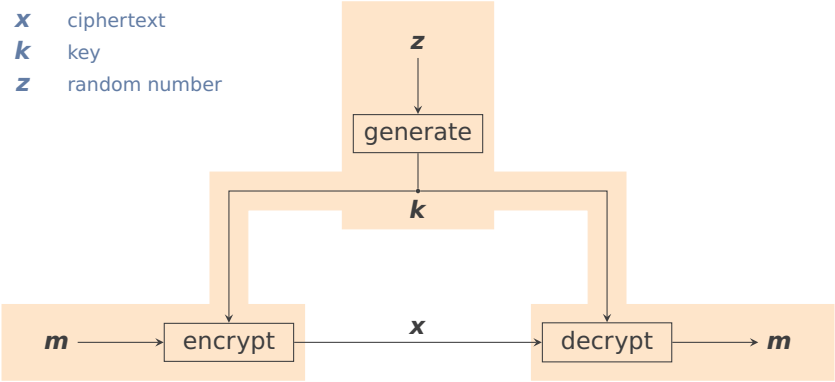
1. **Security of cryptographic functions**
2. Perfect secrecy and the one-time pad
3. Less-than-perfect secrecy

Security of Cryptographic Functions



Symmetric Encryption

m message
x ciphertext
k key
z random number



trusted area

Symmetric Encryption Formalized

A symmetric encryption system is defined by

- message space $\mathcal{M}$,
- ciphertext space $\mathcal{X}$,
- key space $\mathcal{K}$, and
- two deterministic functions with known efficient algorithms

$$\begin{aligned} \text{encrypt}: &\quad \mathcal{M} \times \mathcal{K} \to \mathcal{X} \\ \text{decrypt}: &\quad \mathcal{X} \times \mathcal{K} \to \mathcal{M}. \end{aligned}$$

Correctness

$$\forall (\boldsymbol{m}, \boldsymbol{k}) \in \mathcal{M} \times \mathcal{K} : \text{decrypt}\left(\text{encrypt}(\boldsymbol{m}, \boldsymbol{k}), \boldsymbol{k}\right) = \boldsymbol{m}$$

Agenda

1. Security of cryptographic functions
2. **Perfect secrecy and the one-time pad**
3. Less-than-perfect secrecy

Unconditional Security

Perfect secrecy: Confidentiality of the message is provable without further assumptions on
a) the adversary's computational resources,
b) the hardness of mathematical problems.

Formalization using information theory:

Let $M \sim \mathcal{M}$, $K \overset{\text{unif.}}{\sim} \mathcal{K}$, and $X = \text{encrypt}(M, K)$, then we require

$$H(M \mid X) = H(M) \qquad \longleftarrow \textbf{security}$$

and

$$H(M \mid X, K) = 0 \qquad \longleftarrow \textbf{correctness}$$

$\rightarrow$ **This is the strongest possible definition of confidentiality.**

Shannon 1949

Existence of Unconditionally Secure Encryption

From correctness and deterministic functions:

$$|\mathcal{X}| \geq |\mathcal{M}| \tag{1}$$

$$H(K,X) = H(K, \text{encrypt}(M,K)) = H(K,M) \overset{\text{independent key}}{=} H(K) + H(M) \tag{2}$$

$$= H(\text{decrypt}(X,K),X) = H(M,X) \tag{3}$$

From security:

$$H(M \mid X) = H(M) \overset{\text{chain rule}}{=} H(M,X) - H(X) \overset{(2)}{=} H(K,X) - H(K) \tag{4}$$

$$\overset{(3)}{=} H(K,X) - H(X) = H(K,X) - H(K) \tag{5}$$

$$H(X) = H(K)$$

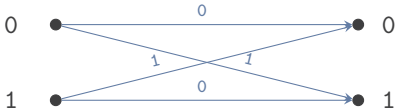
→ **We need at least as many keys as messages:** $|\mathcal{K}| \geq |\mathcal{X}| \overset{(1)}{\geq} |\mathcal{M}|$

One-time Pad (Vernam Cipher)

$n = 1$, $\mathcal{M} = \mathcal{K} = \mathcal{X} = \{0,1\}^n$, $x = \text{encrypt}(m,k) = m \oplus k$

addition modulo 2

$$m \xrightarrow{k} x$$

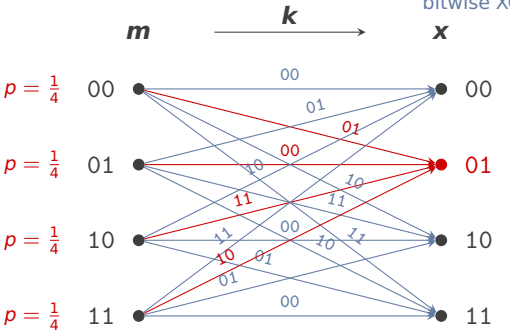


$$P(k=1) = \frac{1}{2} \Rightarrow P(m=1 \mid x=0) = P(m=1 \mid x=1) = P(m=1)$$

One-time Pad (Vernam Cipher)

$n = 2$, $\mathcal{M} = \mathcal{K} = \mathcal{X} = \{0,1\}^n$, $\boldsymbol{x} = \text{encrypt}(\boldsymbol{m}, \boldsymbol{k}) = \boldsymbol{m} \oplus \boldsymbol{k}$

bitwise XOR



perfect secrecy $\Leftrightarrow$ every message is equally likely

Symmetric Encryption with the OTP

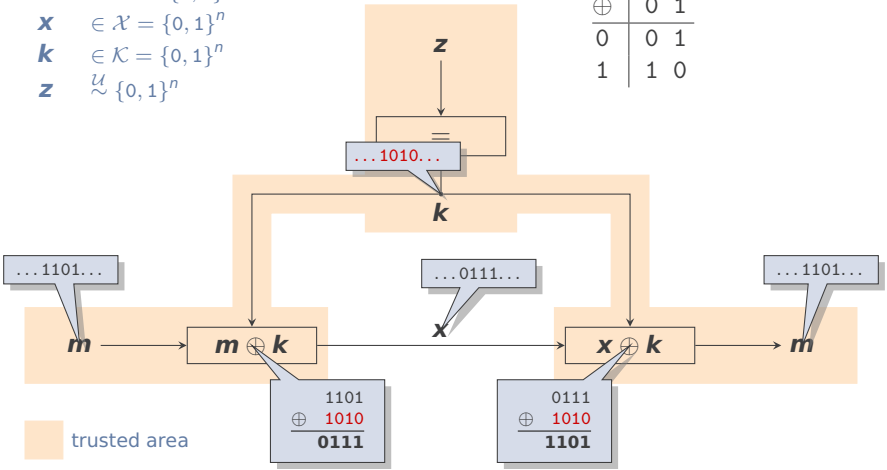
$\boldsymbol{m} \in \mathcal{M} = \{0,1\}^n$

$\boldsymbol{x} \in \mathcal{X} = \{0,1\}^n$

$\boldsymbol{k} \in \mathcal{K} = \{0,1\}^n$

$\boldsymbol{z} \overset{\mathcal{U}}{\sim} \{0,1\}^n$

$\oplus$	0	1
0	0	1
1	1	0



Limitations

The one-time pad is unbreakable if properly used, but . . .

- The key must be truly random.
- **Never reuse key material!**
  $\rightarrow$ Sender and recipient have to stay synchronized.
- Exchanging and protecting huge secret keys is impractical.

  If the key is encrypted with or generated by another cryptosystem, then the overall security is defined by the weakest component.
- No integrity protection: attackers can add message differences.

Agenda

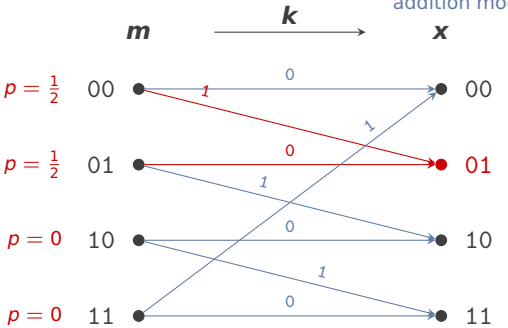
1. Security of cryptographic functions
2. Perfect secrecy and the one-time pad
3. **Less-than-perfect secrecy**

One-time Pad "Economy"

$n = 2$, $\mathcal{M} = \cancel{\mathcal{K}} = \mathcal{X} = \{0,1\}^n$, $\boldsymbol{x} = \text{encrypt}(\boldsymbol{m}, \boldsymbol{k}) = \boldsymbol{m} + \boldsymbol{k}$

$|\mathcal{K}| = 2 < |\mathcal{M}|$

addition modulo 4



imperfect secrecy !

Classification of Attacker Success

(against confidentiality)

Total break	finding the key
Universal break	decryption without knowing the key
Semantic break	learning a “bit” about the message $\rightarrow$ ability to distinguish ciphertexts
Existential break	success for individual messages selective $\leftrightarrow$ random
Distinguisher	between ciphertext and random bits

Classification of Attacker Capabilities

(against confidentiality)

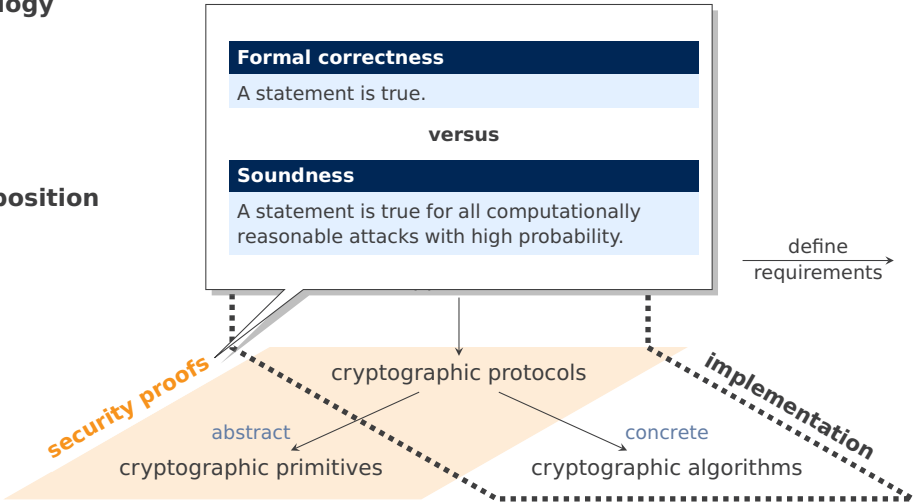
Ciphertext only	find plaintext	CO
Known plaintext	find other plaintext or key	KPA
Chosen plaintext	distinguish ciphertexts $\rightarrow$ access to encryption oracle	CPA
Chosen ciphertext	distinguish ciphertexts $\rightarrow$ access to decryption oracle	CCA
Adaptively chosen ciphertext	distinguish other ciphertexts $\rightarrow$ mix choice and oracle queries	CCA2
Related key	find plaintext or key $\rightarrow$ perform operations on key	

Note: most sources formalizing these notions assume indeterministic encryption. We shall come to this later.

Cryptology

Ontology

Composition



Two Ways of Formalizing “Soundness”

Concrete security

A primitive is (t, ϵ)-secure if the advantage of every attacker running time t is at most ϵ.

Example

A perfect four-digit lock is $(10, 1/1000)$-secure; or $(5000, 1/2)$-secure; ...

Asymptotic security

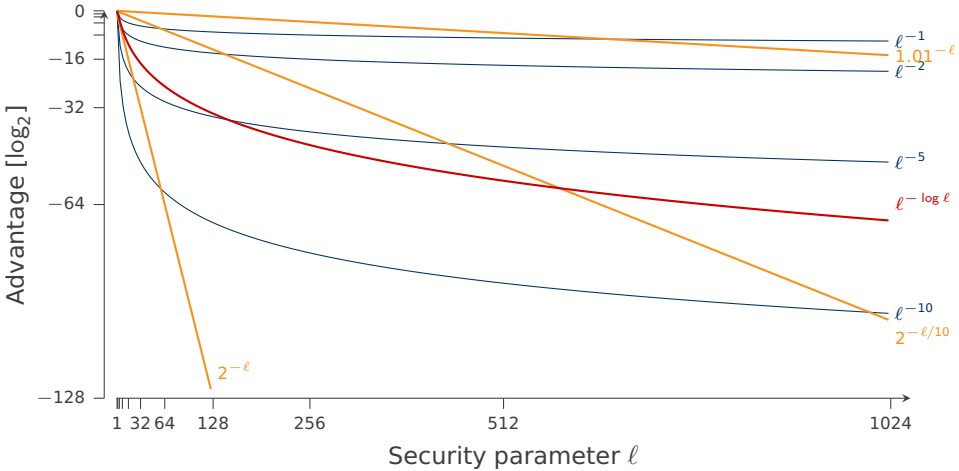
The advantage is a *negligible function* of a *security parameter* ℓ (e. g., key length) for every attacker running in polynomial (in ℓ) time.

A function $f : \mathbb{N} \to [0, 1]$ is called *negligible* if for every polynomial p, there is $\ell_0 \in \mathbb{N}$ such that $f(\ell) < p(\ell)^{-1}$ for every $\ell > \ell_0$. — “decreases faster than every polynomial”

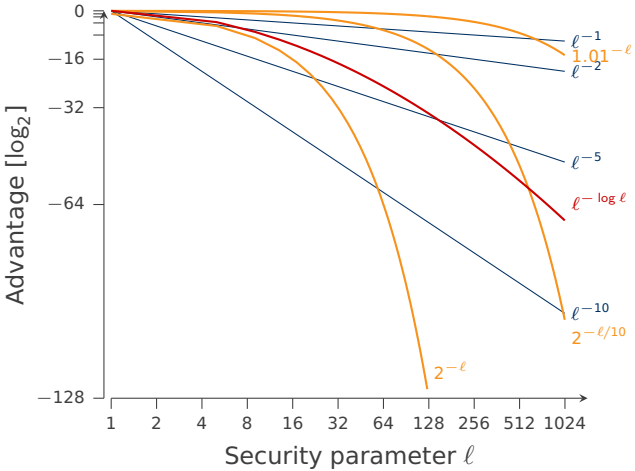
Examples

$\ell \mapsto 2^{-\ell}$; $\ell \mapsto 1.001^{-\ell}$; $\ell \mapsto 2^{-\ell/10}$; $\ell \mapsto \ell^{-\log \ell}$

Negligible Functions



Negligible Functions (log–log)



Summary

Today we have learned . . .

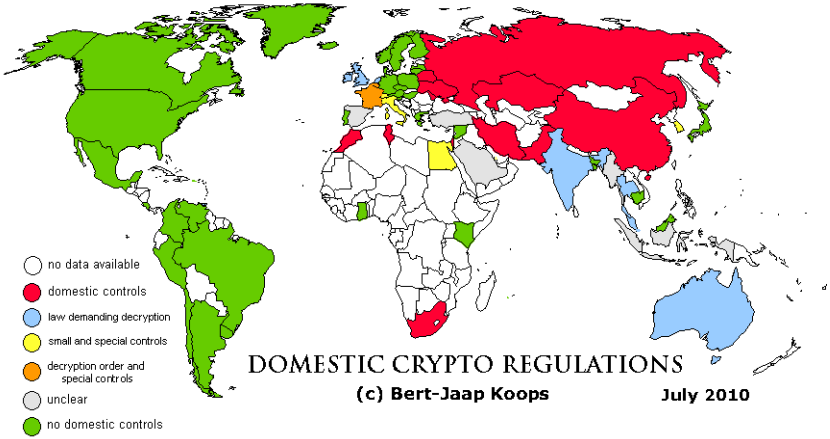
1. how we qualify confidence in security propositions;
2. how information theory helps us to define confidentiality;
3. some terminology to precisely talk about less-than-perfect confidentiality.

Learning for life

We know how to construct the best encryption system possible.

And we can memorize its design !

Cryptography and Freedom



Source: http://www.cryptolaw.org/

Syllabus

Summer Term 2026

	05.03.26	1.	Introduction, basic terminology	
	12.03.26	2.	Probability, information, and coding theory for cryptography	
	19.03.26	3.	Symmetric encryption, perfect secrecy	
Tue	24.03.26	4.	Block ciphers	HSB 9
	16.04.26	5.	Cryptanalysis of classical ciphers	
	23.04.26	6.	Number theory for cryptography	
	30.04.26	7.	Computational complexity for cryptography	
	07.05.26	8.	Asymmetric encryption	
	21.05.26	9.	Message authentication	
	28.05.26	10.	Elliptic curve cryptography	
	11.06.26	11.	Commitments, zero-knowledge, and secret sharing	
Tue	23.06.26	12.	Q & A for the exam	HSB 9
	25.06.26		**Written exam**	

Last update: March 18, 2026